The Mathematical Tools of Quantum Mechanics (Cohen–Tannoudji, Ch. II) Section A. One-particle wave function space

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Outline

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- B. State space and Dirac notation

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- C. Representations in the state space
- D. Eigenvalue equations and observables
- E. Two key examples: $\{|\boldsymbol{r}\rangle\}$ and $\{\ |\boldsymbol{p}\rangle\}$; operators $\boldsymbol{R}$, $\boldsymbol{P}$
- F. Tensor products and applications

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- Normalization: $\int_{\mathbb{R}^3} |\psi(\boldsymbol{r}, t)|^2 \, \mathrm{d}^3 r = 1$.
- This leads us to square-integrable functions (where the integral converges): $\psi \in \mathcal{L}^2(\mathbb{R}^3)$, a Hilbert space.

$\mathcal{F}$ as a vector space

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- We call $\mathcal{F}$ the set of these specific wave functions ($\mathcal{F}$ is a subspace of $\mathcal{L}^2$)
- Closure under linear combinations: if $\psi_1, \psi_2 \in \mathcal{F}$ and $\alpha, \beta \in \mathbb{C}$, then $\alpha\psi_1 + \beta\psi_2 \in \mathcal{F}$.

Scalar product and norm

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  - Linear in second argument (antilinear in first)

$$(\varphi, \lambda_1\psi_1 + \lambda_2\psi_2) = \lambda_1(\varphi, \psi_1) + \lambda_2(\varphi, \psi_2)$$
$$(\lambda_1\varphi_1 + \lambda_2\varphi_2, \psi) = \lambda_1^*(\varphi_1, \psi) + \lambda_2^*(\varphi_2, \psi)$$

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  - Positive definiteness

$$(\psi, \psi) = \int d^3 r\, |\psi(\mathbf{r})|^2$$

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- Schwarz inequality

$$|(\psi_1, \psi_2)| \leq \sqrt{(\psi_1, \psi_1)} \sqrt{(\psi_2, \psi_2)}$$

Becomes an equality if and only if the two functions ψ_1 and ψ_2 are proportional

Linear operators on wave functions

- A linear operator A is a mathematical entity which, with every function $\psi(\mathbf{r}) \in \mathcal{F}$, associates another function $\psi'(\mathbf{r})$:

$$\psi'(\mathbf{r}) = A\psi(\mathbf{r})$$

$$A\big[\lambda_1\psi_1(\mathbf{r}) + \lambda_2\psi_2(\mathbf{r})\big] = \lambda_1 A\psi_1(\mathbf{r}) + \lambda_2 A\psi_2(\mathbf{r})$$

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  - Derivative $D_x\psi = \partial_x\psi$.

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$$\begin{aligned}\big[X,\ D_x\big]\,\psi(x) &= X(\partial_x\psi) - \partial_x(x\psi) \\ &= x\,\partial_x\psi - (\psi + x\,\partial_x\psi) \\ &= -\psi(x)\end{aligned}$$

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Thus $[X,\ D_x] = -\mathbb{I}$.

Discrete orthonormal bases $\{u_i(\boldsymbol{r})\}$

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- Components are obtained as:

$$c_i = (u_i, \psi) = \int u_i^*(\boldsymbol{r}) \, \psi(\boldsymbol{r}) \, d^3r.$$

Scalar product in components

Let $\phi(\boldsymbol{r})$ and $\psi(\boldsymbol{r})$ be two wave functions which can be expanded as:

$$\phi(\boldsymbol{r}) = \sum_i b_i u_i(\boldsymbol{r}) \qquad \psi(\boldsymbol{r}) = \sum_j c_j u_j(\boldsymbol{r})$$

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- In particular:

$$(\psi, \psi) = \|\psi\|^2 = \sum_i |c_i|^2$$

Closure relation (discrete basis)

If $\{u_i(\boldsymbol{r})\}$ is a basis, there exists an expansion for every function $\psi(\boldsymbol{r})$ such that:

$$\psi(\mathbf{r}) = \sum_i c_i\, u_i(\mathbf{r}) = \sum_i (u_i, \psi)\, u_i(\mathbf{r}) \tag{1}$$

$$= \sum_i \left[\int d^3r'\, u_i^*(\mathbf{r}')\, \psi(\mathbf{r}') \right] u_i(\mathbf{r}) \tag{2}$$

$$= \int d^3r'\, \psi(\mathbf{r}') \left[\sum_i u_i(\mathbf{r})\, u_i^*(\mathbf{r}') \right] \tag{3}$$

- If $\psi(\mathbf{r}) = \int d^3r'\, \psi(\mathbf{r}') F(\mathbf{r}, \mathbf{r}')$, then $\sum_i u_i(\boldsymbol{r})\, u_i^*(\boldsymbol{r}\,') = \delta(\boldsymbol{r} - \boldsymbol{r}\,')$.

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- Guarantees unique expansion in the chosen basis.

"Bases" not belonging to $\mathcal{F}$: plane waves

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  Remember the Fourier transform $\bar{\psi}(p)$ of $\psi(x)$:

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- Coefficients (FT relations): $\bar{\psi}(p) = \int v_p^*(x)\, \psi(x)\, \mathrm{d}x.$

Orthogonality and closure (plane waves)

Using the relation $\frac{1}{2\pi}\int \mathrm{d}k\, e^{iku} = \delta(u)$

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- Closure relation: $\int \mathrm{d}p\, v_p(x)\, v_p^*(x') = \frac{1}{2\pi}\int \frac{\mathrm{d}p}{\hbar}\, e^{\mathrm{i}\frac{p}{\hbar}(x-x')} = \delta(x-x')$

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- Orthonormalization:
  $(v_p, v_{p'}) = \int \mathrm{d}x\, v_p^*(x)\, v_{p'}(x) = \frac{1}{2\pi}\int \frac{\mathrm{d}x}{\hbar}\, e^{\mathrm{i}\frac{x}{\hbar}(p-p')} = \delta(p-p').$

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- Norm: $\|\psi\|^2 = \int |\psi(\boldsymbol{p})|^2\, \mathrm{d}^3 p$.

Continuous "basis": delta functions

Let's introduce a set of function of x, $\{\xi_{x_0}(x)\}$, labeled by continuous index x_0, defined by $\xi_{x_0}(x) = \delta(x - x_0)$. $\xi_{x_0}(x) \notin \mathcal{F}$.

For every function $\psi(x) \in \mathcal{F}$ we can write:

$$\psi(x) = \int \mathrm{d}x_0 \, \psi(x_0) \, \delta(x - x_0) = \int \mathrm{d}x_0 \, \psi(x_0) \, \xi_{x_0}(x) \tag{4}$$

$$\psi(x_0) = \int \mathrm{d}x \, \delta(x_0 - x) \, \psi(x) = \int \mathrm{d}x \, \xi^*_{x_0}(x) \, \psi(x) \tag{5}$$

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$\psi(x_0)$ is therefore the equivalent to c_1: represent the components of $\psi(x)$ in the chosen basis

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  IMPORTANT NOTE: A physical state must always correspond to a square-integrable wave function. In no case can $v_{\boldsymbol{p}(\boldsymbol{r})}$ or $\xi_{\boldsymbol{r_0}}(\boldsymbol{r})$ represent the state of a particle. They are intermediaries.

General continuous "orthonormal" bases

- A continuous orthonormal basis is a set of functions of $\boldsymbol{r}$, $\{w_\alpha(\boldsymbol{r})\}$, labeled by a continuous index α, which satisfy orthonormalization and closure relations:

$$(w_\alpha, w_{\alpha'}) = \int \mathrm{d}^3 r \; w_\alpha^*(\boldsymbol{r}) w_{\alpha'}(\boldsymbol{r}) = \delta(\alpha - \alpha') \tag{6}$$

$$\int w_\alpha(\boldsymbol{r}) w_\alpha^*(\boldsymbol{r}\,') \, \mathrm{d}\alpha = \delta(\boldsymbol{r} - \boldsymbol{r}\,'). \tag{7}$$

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- Components of the wave function (unique expansion):

$$\psi(\boldsymbol{r}) = \int \mathrm{d}^3 r' \, \psi(\boldsymbol{r}') \, \delta(\boldsymbol{r} - \boldsymbol{r}') = \int \mathrm{d}\alpha \left[\int \mathrm{d}^3 r' \, w_\alpha^*(\boldsymbol{r}') \, \psi(\boldsymbol{r}') \right] w_\alpha(\boldsymbol{r}) \tag{8}$$

$$= \int c(\alpha) \, w_\alpha(\boldsymbol{r}) \, \mathrm{d}\alpha, \quad \text{with } c(\alpha) = \int \mathrm{d}^3 r' \, w_\alpha^*(\boldsymbol{r}') \, \psi(\boldsymbol{r}'). \tag{9}$$